Lecture 8

Thursday, 14 September 2023 10:34 AM

Propane Dehydrogenation [Pd-Zn catalyst
↓
surface monolayer

C-C activation is more favorable (higher equilibrium conversion at lower temp.)

Structure sensitive vs. Structure insensitive
↓ (Dehydro)
reaction at vertex/edge

C-C bonds are broken at np. surfaces (not edges) → Structure sensitive

Zn, Ga, In, Sn, Cr → other alloys used w/ Pt

Pd is awful at PDH → Pd FCC

→ β_1-PdZn (Intermetallic) phase } has specific atomic distance (b/w atoms) → This has isolated Pd sites

↓

So thought that it would be good for catalysis } Looking at diff. alloys on the phase diagram chart

↓

Make core-shell structures a lot

EXAFS → coordination no.
Δ bond distance b/w atoms

Alloy formation as function of time
(when is the β phase formed)

Electronic Effect:

d-band center used as descriptor for reactivity